

Session 1.1: Course Orientation & The Fundamentals of Platform Businesses

Linear value chains vs. two-sided ecosystem orchestration, transaction cost economics, and the asset-light growth model.

PART 1 (LECTURE)
60 Minutes

PART 2 (LAB PRACTICE)
60 Minutes

PART 3 (EXECUTIVE MEMO)
60 Minutes

CORE METHOD
SEC 10-K Reconciliation

Learning Objectives & Core Vocabulary

Clear definitions for students with zero prior platform economics background.

CORE COMPETENCIES

Goals

What You Will Master Today

- **Deconstruct Value Models:** Distinguish linear pipeline production from multi-sided ecosystem orchestration.
- **Apply Economic Theory:** Explain how digital software protocols eliminate Coasean transaction friction.
- **Audit Financial Statements:** Calculate fixed asset intensity and labor leverage from SEC 10-K regulatory filings.
- **Formulate Strategy:** Advise executive management on navigating legacy-to-platform channel conflict.

FOUNDATIONAL TERMS

Vocabulary

Five Essential Concepts

- **Pipeline Business:** Linear step-by-step value chain where the firm owns inventory and controls inputs to outputs.
- **Platform Business:** Digital infrastructure that matches external producers and consumers to enable direct exchanges.
- **Intermediation:** Automated coordination of transactions without taking balance-sheet ownership of physical assets.
- **Asset-Light Model:** Scaling revenue rapidly by substituting software code for capital expenditure and physical property.
- **Operational Leverage:** Growing transactions exponentially while keeping corporate headcount and fixed overhead flat.

Theory 1: Linear Pipelines vs. Platform Ecosystems

Van Alstyne, Parker, and Choudary (2016) : Pipelines, Platforms, and the New Rules of Strategy

Strategic Dimension	Linear Pipeline Model	Multi-Sided Platform Model
<i>Value Architecture</i>	Linear supply chain: Supplier → Manufacturer → Distributor → Consumer.	Ecosystem matrix: Orchestrates external exchanges between producers and consumers.
<i>Asset Base & PP&E</i>	Heavy property, plant, and equipment (PP&E) owned directly on balance sheet.	Asset-light software infrastructure; physical assets owned by external ecosystem participants.
<i>Scale Mechanism</i>	Supply economies of scale: Lowering per-unit fixed costs through physical plant volume.	Demand economies of scale: Value accelerates via network effects as participant count grows.
<i>Growth Bottleneck</i>	Capital expenditure (CapEx), physical construction lead times, and factory capacity.	Algorithmic match quality, user trust, liquidity density, and transaction safety.

STRATEGIC TAKEAWAY FOR BUSINESS LEADERS

Pipelines compete by optimizing internal processes and controlling proprietary resources. Platforms compete by governing external interactions and orchestrating vibrant community ecosystems.

Theory 2: Transaction Cost Economics & Digital Networks

Ronald Coase (1937) The Nature of the Firm & Oliver Williamson (1981) The Economics of Organization

FRICTION 1

Discovery

Search Costs

The time, labor, and money required for buyers and sellers to discover compatible trading partners in an open market.

- **Legacy World:** Travel agencies, printed catalogs, local physical directories.
- **Digital Platform:** Algorithmic real-time indexing, search filters, instant price discovery.

FRICTION 2

Agreement

Contracting Costs

The legal, negotiation, and escrow expenses necessary to reach binding agreements and protect counterparty interests.

- **Legacy World:** Individual bespoke contracts, lawyers, notarized guarantees.
- **Digital Platform:** Standardized terms of service, programmatic insurance policies, integrated escrow.

FRICTION 3

Fulfillment

Coordination Costs

The operational overhead required to enforce terms, verify performance, and resolve bilateral transaction disputes.

- **Legacy World:** On-site human inspection, customer service hotlines, court arbitration.
- **Digital Platform:** Two-way verified reputation systems, automated refund protocols, telemetry tracking.

THE COASEAN DIGITAL SHIFT

When digital software protocols reduce search, contracting, and coordination costs toward zero, transactions that previously required massive integrated corporations can now occur efficiently across open decentralized networks.

Showdown 1: Traditional Publishing vs. Amazon KDP

Physical gatekeeping and batch print runs vs. algorithmic print-on-demand infrastructure.

TRADITIONAL PUBLISHER (PIPELINE MODEL)

Legacy Gatekeeping

Editorial Selection & Physical Batches

- **Editorial Gatekeepers:** Review boards reject 99% of unsolicited manuscripts, filtering content to minimize printing downside risk.
- **Inventory Risk:** Requires minimum batch runs of 5,000 physical copies; unsold books must be warehoused or pulped at steep loss.
- **Supply Chain Lead Time:** Requires 12 to 18 months from manuscript completion to physical bookstore distribution.
- **Value Distribution:** Authors receive 10% to 15% royalty; publisher absorbs retail discounts and logistics overhead.



AMAZON KDP (PLATFORM MODEL)

Ecosystem Orchestrator



Algorithmic Print-on-Demand

- **Open Access Onboarding:** Any global author uploads a manuscript file; the title is cataloged worldwide within 24 hours.
- **Zero Inventory Carrying:** Automated Print-on-Demand (POD) systems print single copies only after a verified customer purchase.
- **Algorithmic Discovery:** Community customer reviews and algorithmic recommendation replace corporate editorial taste.
- **Value Distribution:** Authors capture up to 70% royalties; Amazon monetizes cloud infrastructure and marketplace reach.

CORE STRATEGIC DISPARITY

Traditional publishing constrains supply to mitigate physical inventory risk. Amazon KDP expands supply infinitely by driving marginal inventory costs to zero, monetizing the long-tail curve.

Showdown 2: Commercial Banking vs. Stripe Payments

Physical marble branch networks vs. seven lines of programmable cloud payment code.

COMMERCIAL RETAIL BANK (PIPELINE MODEL)

Physical Branches

Branch Leases & Manual Underwriting

- **Physical Branch Footprint:** Sinks billions into high-rent street corners, bulletproof glass, human tellers, and armored cars.
- **Manual Merchant Underwriting:** Merchant terminal accounts require paper applications, credit checks, and 2-3 weeks delay.
- **Legacy Clearing Batches:** Transactions settle in overnight batches across fragmented legacy clearinghouses.
- **Geographic Enclosure:** Services constrained by national banking charters, physical regional borders, and local currency silos.

S STRIPE INC. (PLATFORM MODEL)

Developer API Platform

S Seven Lines of Code

- **Developer-First Integration:** Software engineers paste seven lines of JavaScript to accept credit cards globally in minutes.
- **Algorithmic Risk Scoring:** Machine learning models (Stripe Radar) evaluate multi-factor fraud risk on billions of data points in milliseconds.
- **Global Currency Protocol:** Seamless programmatic clearing across 135+ currencies without opening physical bank accounts.
- **Software Scalability:** Processes over \$1 trillion in annual transaction volume with a lean, world-class engineering team.

CORE STRATEGIC DISPARITY

Commercial banks compete through capital-intensive real estate and human gatekeepers. Stripe competes as developer-first cloud infrastructure, transforming financial plumbing into programmable software.

Case Showdown: Marriott International vs. Airbnb Inc.

Empirical confrontation between physical asset ownership and software orchestration (2015-2023).

 MARRIOTT INTERNATIONAL (PIPELINE GIANT)

PP&E Heavy Model

 Real Estate & Hotel Leases

- **Physical Expansion:** Invested over \$13 billion in cash and debt to acquire Starwood Hotels (2016), absorbing hotel leases and renovations.
- **Active Room Inventory:** ~1.6 Million rooms across 30 legacy brands; adding 10,000 rooms requires hundreds of millions in CapEx.
- **Balance Sheet PP&E:** Carries \$1,861M in Net PP&E (~7.4% of total assets) plus heavy long-term operating lease liabilities.
- **Labor Productivity:** Employs ~411,000 managed associates; generates ~\$57,700 in annual revenue per worker.

 AIRBNB INC. (PLATFORM ORCHESTRATOR)

Asset-Light Network

 Software Orchestration

- **Zero-CapEx Expansion:** Recruited independent hosts globally via algorithmic search optimization, reaching 7.7 million listings with zero land purchases.
- **Global Inventory Scale:** Over 7.7 Million active listings across 100,000 cities, outscaling Marriott's room count by 4.8x.
- **Asset-Light Balance Sheet:** Net PP&E is under 0.8% of assets (\$154M); corporate infrastructure is hosted entirely in the cloud.
- **Supercharged Labor Leverage:** 6,889 corporate employees generate ~\$1,438,000 in revenue per worker (25x Marriott's productivity).

AUDITED SEC 10-K FINDING

By substituting software protocols for physical property ownership, Airbnb achieves 4.8x greater listing scale and 25x higher labor productivity than Marriott while carrying negligible fixed asset risk.

Lab Transition: 120-Minute Hands-On Execution

Shifting from lecture observation to audited data manipulation and executive decision-making.

EXERCISE 1: FOUNDATIONAL PRACTICE

60 Min Lab

Audited SEC 10-K Financial Lab

Step-by-step spreadsheet model to compute and verify asset intensity metrics:

- **Dataset:** Open `data/session_1_1_asset_intensity.csv` in Excel or Google Sheets.
- **Step 3 (PP&E Ratio):** In cell `J2`, enter formula `=D2/C2*100` (Net PP&E to Assets).
- **Step 4 (Labor Productivity):** In cell `K2`, enter formula `=(E2*1000000)/G2` (Revenue per Worker).
- **Audited Finding:** Quantify the 25x labor leverage advantage of digital platform orchestration.

EXERCISE 2+: ADVANCED PROBLEM SOLVING

60 Min Memo

Executive Dilemma & Board Strategy Memo

Resolve Marriott's real-world corporate challenge with "Homes & Villas":

- **The Dilemma:** Traditional pipeline hotel operators face high CapEx; can Marriott operate a marketplace without cannibalizing franchisees?
- **Key Tradeoffs:** Premium brand quality control vs. rapid host onboarding; franchise owner backlash.
- **Student Deliverable:** Draft a structured 3-bullet decision brief for the Board of Directors: Build internally or acquire an existing rental platform?